Tell It Like It (Usually) Is(n't): Exploring Newsworthiness and (A)typical Tool Mentions on Reddit

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Abstract

Language is often used as a channel for newsworthy or atypical information. We investigate whether online discourse contains a prevalence of descriptions of atypical situations or typical situations. We focus on mentions of tool use, i.e., using an object as a tool to complete a task. Additionally, we consider whether tools mentioned more often in positive contexts (e.g., "with TOOL") differ in typicality from tools mentioned in negative contexts (e.g., "without TOOL"). First, we extract mentions of tool use from Reddit. Then, we obtain human typicality judgements for each tool and task. We find that participants rate these tool-use scenarios overall as more atypical than typical. Further, when tools occur more often in positive constructions, they are rated as less typical, whilst when they appear more often in negative constructions, they are rated as more typical. We discuss the implications of these findings for NLP systems.

Keywords: language processing; typicality; atypicality; newsworthiness; tool/instrument use; Reddit; human judgments

Introduction

If speakers use language to describe their world, one might expect frequent mentions of typical, real-world situations (`TELL IT LIKE IT IS`). Yet an essential function of language is to convey non-inferable, newsworthy content, especially when speakers make rational decisions about syntactically optional elements (`TELL IT LIKE IT ISN'T`). Thus, communication often prioritises newsworthy content. A conversational contribution is newsworthy if it is informative and important enough to be reported (cf. Grice, 1975). The tendency to produce mainly newsworthy content in conversation aligns with the Gricean maxim of Quantity, which includes two submaxims: "1. Make your contribution as informative as is required (for the current purposes of the exchange). 2. Do not make your contribution more informative than is required" (Grice, 1975, p. 45).

Beyond pragmatic theory, studies have demonstrated that the notions of predictability and (a)typicality play a key role in language production and comprehension (Janczura & Nelson, 1999; Mitchell et al., 2013). In production, speakers prefer atypicality; that is, unpredictable and implausible content worthy of being conveyed (Brown & Dell, 1987; Mitchell et al., 2013). For instance, atypical instruments are more likely to be mentioned, as are modifiers of atypical properties (e.g., an icepick rather than a knife for stabbing in Brown & Dell, 1987; blue rather than red strawberry in Sedivy, 2003; wool rather than ceramic bowl in Mitchell et al., 2013). On the other hand, in comprehension, findings are less consistent. Some studies suggest that comprehenders prefer typical information that is in keeping with their world knowledge and therefore yields easier processing (Degen et al., 2020). The preference for typical information is also influenced by the comprehension difficulties caused by negation (Just & Carpenter, 1971; Zwaan, 2012). However, recent studies suggest a comprehension bias towards atypical information, showing ease of processing for sentences about implausible situations (Rohde et al., 2021). These conflicting results highlight the importance of continuing to examine the role of (a)typicality in language.

While previous research has predominantly relied on language production in psycholinguistic experiments to understand the role of (a)typicality, we examine online data to better understand how (a)typicality influences communicative choices in natural environments. Specifically, in this study we focus on descriptions of (a)typical tool use as such descriptions provide an effective way to explore how real-world knowledge and situational context shape communication. We use the word "tool" to mean an object which enables a certain task; however, other terms, such as "instrument", are also used in the literature for this purpose. To study (a)typicality in naturally occurring productions, we analyse Reddit[1] data, examining the preference for atypical and newsworthy content in online communication.

Furthermore, we distinguish between two scenarios: where the tool use has been mentioned in the context of a negation word or phrase (negative construction), and where there is no negation (positive construction), e.g., "painting a picture **without** a brush" and "eating soup **with** a fork" respectively. This distinction is crucial because, under our hypothesis that the Reddit data reflects people's propensity to `TELL IT LIKE IT ISN'T`, negative constructions would be expected to highlight the unexpected *absence* of typical tools, whereas positive constructions would highlight the unexpected *presence* of atypical tools used in a novel or surprising way.

In order to determine whether a particular mention of a tool-use scenario is typical or atypical, we collect judgements of typicality from human participants. We predict, in line with previous findings on the importance of newsworthy content, that:

1. Tool-use scenarios from Reddit will be rated on average as

[1] `https://www.reddit.com/`

more atypical than typical by human participants.

2. Tool-use scenarios that appeared more often on Reddit in a negative construction (e.g., "without TOOL") will be rated as more typical than those mentioned more often in positive constructions (e.g., "with TOOL").

Methods

Data

To evaluate the typicality of tool-use mentions in text corpora, we used data from Reddit. Reddit is an online social network consisting of informal discussion threads, without the length restrictions that other platforms impose. This makes Reddit a good source for naturalistic, informal language data. Our study uses data from comments, excluding submissions (initial post content). Comments tend to be more conversational, while submissions are often either questions posed to users, long-winded anecdotes, or non-textual content such as images. We collected 4 months of Reddit comments (March 2021–June 2021) from Pushshift,[2] totalling over 88 billion comments. We first filtered the comments automatically and then manually. The result was a set of VERB-DOBJ-TOOL triplets, called action/instrument combinations in Rohde et al. (2021) (e.g., `eat-soup-dessert_spoon`, `cut-furniture-chainsaw`, `catch-bugs-net`).

Automatic Filtering

Syntactic Patterns Following Rohde et al. (2021), we only included sentences which described an **action** being performed on a **direct object** using a **tool**. We only considered triplets where VERB was one of the following (half of which are from Rohde et al. (2021) and half are new): *cut, chop, dig, brush, clean, repair, mend, sew, knit, transport, drain, wash, write, eat, paint, sweep, stir, spread, heat, catch*.[3]

Using spaCy,[4] we automatically parsed the comments into sentences, lemmatized the words, and tagged the words with their part-of-speech tag. We then used spaCy to select sentences that matched one of four syntactic patterns, where USE is any form of the word "use":[5] VERB DOBJ with/using TOOL; USE TOOL to VERB DOBJ; VERB DOBJ without/using no TOOL; not/never USE TOOL to VERB DOBJ.

This procedure yielded 27,301 sentences. Here are some examples:

(a) **Eating sushi with a fork** cause there were no available chopsticks

(b) I **eat cheese with coffee**, it tastes good ngl

[2]This data was publicly available in 2021 from https://pushshift.io but access has since been restricted.

[3]We reduced the search space in this way due to time considerations. This enabled us to only consider verbs which we knew to involve tool use, but ideally we would have considered an open-ended set.

[4]https://spacy.io/

[5]The DOBJ and TOOL are both noun phrases that can optionally include either a determiner, a pronoun, or a numeral, and any number of adjectives.

(c) However, you can always **wash your hair without shampoo**

(d) Let's **not use the internet to spread false information**

While (a) is describing tool use, (b) is clearly not. At the same time, (c) and (d) are not easily classified. Since pattern matching was not sufficient to remove irrelevant sentences, we performed another round of automatic filtering as follows.

Miscellaneous We filtered out offensive sentences using a manually constructed list of obscene/offensive words. We also removed sentences which were repeated many times in the corpus and appeared to be automatically generated by a bot. Finally, we removed any sentences where the direct object or tool was not a concrete noun. Concreteness ratings came from Brysbaert et al. (2014), a collection of around 40,000 English nouns which were rated by human participants on Amazon Turk on a scale of 1 (abstract) to 5 (concrete). We considered a noun to be concrete if its mean concreteness rating was greater than 4. Nouns that were not included in the concreteness ratings were also filtered out. This excluded sentences such as "Let's not use the internet to spread false information". This round of filtering resulted in 9,621 sentences. Some example sentences are shown in Table 1.

Manual Filtering

To ensure the remaining sentences only contained real descriptions of tool use, the authors manually annotated a random sub-sample of the data. We judged triplets rather than full sentences, since this is the same form of stimuli we intended to use in the Participant Study. Triplets were used so that the judgements are about whether a given tool is atypical for a given task in a purely conceptual way, without further linguistic context. To construct the target set of triplets, subsamples of 250 positive and 250 negative VERB-DOBJ-TOOL triplets were randomly selected from our sentences, excluding duplicates and considering only triplets where the VERB-DOBJ combination occurred in the data more than once.

We manually filtered out any triplets where the TOOL was not enabling the action, such as in `eat-burger-cheese` where `cheese` is accompanying the burger but not enabling the eating of the burger. We also removed triplets containing brand names (since they may not be understood by all the participants), obscenities or offensive content which may have been missed in automatic filtering, non-human actions (e.g., `eat-pear-little_paws`), slang/culture-specific/idiomatic expressions (e.g., `paint-women-one_broad_brush`), and anything making clear reference to a previous mention (e.g., `clean-hands-same_water`, presumably referring to an action done first and then cleaning something else with the same water).

Some of the triplets were judged by more than one author. We removed any triplets that none of us considered to be an example of tool use. This resulted in 499 triplets (see the full set on OSF).

Table 1: Examples of tool-use mentions extracted from Reddit, for both positive and negative constructions.

	Pattern	Example
Positive	VERB DOBJ with/using TOOL	No, I **eat soup with a dessert spoon**
	USE TOOL to VERB DOBJ	You can **use a small electric chainsaw to cut a piece of furniture** up
Negative	VERB DOBJ without/using no TOOL	Well, if you **catch bugs without a net**, they get squished in your hand
	not/never USE TOOL to VERB DOBJ	Do**n't use chopsticks to eat pizza**, its undignified

Participant Study

We conducted a human participant study to collect typicality judgments regarding the use of different tools for different actions, as found in the Reddit data.

Participants 206 participants were recruited via Prolific.[6] Participants were aged 18 or over, self-reported native speakers of English, and residents of the US. Participants were each compensated REDACTED FOR REVIEW [AMOUNT IN COMPLIANCE WITH PROLIFIC HOURLY RATE] for their participation, which lasted approximately 10 minutes, and gave informed consent before commencing the task.

Materials The stimuli were presented online using jsPsych (de Leeuw, 2015), and consisted of the triplets of VERB-DOBJ-TOOL, extracted as described above. The stimuli did not indicate whether the triplet was originally used in a positive or negative construction.

Each word in the triplet was presented alone – wrapped in [ ] brackets, to help designate which word belonged to which part of the sequence – in the standard order found in English, with no prepositions, pronouns, or other function words. Content words directly describing a tool (e.g., specific colours) were retained to prevent any potential change in meaning, and were contained within the [ ] of the TOOL itself. The text of the TOOL was presented in a different coloured font, in bold text, and with an underline. The typographical choices aimed to make the contrast as accessible as possible for participants with atypical colour vision.

Each triplet was presented along with a brief reminder of the instructions. Figure 1 shows an example test trial. A total of 499 triplets were used in the experiment.

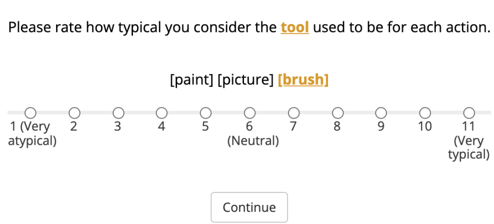

Figure 1: Example test stimulus, instructions, and Likert scale.

[6] This study was granted ethical approval by REDACTED FOR REVIEW.

Procedure Participants were exposed to a short (one-trial) training phase to familiarise them with the experimental procedure. They were given instructions prior to the training phase, and then during the training phase were again reminded that "for each question, you will see the [action] [what the action is being done to] [what the action is being done with]". The training stimulus was a typical example of a tool used for an action: [clean] [floor] [vacuum]. They were asked to rate the typicality of the tool used in the action on a 1 (very atypical) to 11 (very typical) Likert scale. Only the two extremes of the Likert scale were labelled. Participants indicated their typicality rating by selecting one of the 11 points on the scale. An 11-point scale allowed us to treat ratings as quasi-continuous in statistical analysis (Wu & Leung, 2017).

The testing trials were identical to the training trial, with participants asked to rate the typicality of the TOOL in a series of triplets. 20 participants rated each triplet group, with the exception of group 4 of the triplets which was seen by 21 participants. Thus, each triplet was seen at least 20 times, up to a total of 21 times. Each trial was required. The triplets were randomly divided into 9 groups of 50 and 1 group of 49, each participant seeing one of the 10 possible groups.

Participants were required to undertake three attention checks randomly interspersed throughout the main set of trials, asking them to select a specific number (1, 6 or 11) on a Likert scale. Participants who failed to correctly respond to 2/3 attention trials were excluded; no participants failed to meet this criteria. 2 participants were excluded on the basis of failing to complete the experiment, though 1 participant later re-took and completed the study, and their final data was included. 4 participants took part in a small pilot run of the experiment in order to check functionality; due to changes made as a result of this pilot run, they were also excluded.

Results

In total, the data from 201 participants was analysed. Data was analysed in R, version 4.1 (R Core Team, 2021) using the lme4 package (Bates et al., 2015) to fit linear mixed-effects models. All models included a by-participant, by-verb, by-direct object, and by-tool random intercept, to account for subject and item-specific variance. To determine the significance of the fixed effects of interest, models with differing or no mixed effects were compared using Likelihood Ratio tests to obtain the χ^2 statistic, thus comparing model fit. P-values were also obtained from t-tests using Satterthwaite approximation when fitting the models.

Recall that our first prediction was that tool-use mentions on Reddit would be rated as more atypical than typical. To investigate this, we first fit a baseline model with participant typicality rating as the dependent variable and only an intercept term. The intercept-only model, however, provides support for our first prediction that, overall, people rate tool-use scenarios from Reddit data as more atypical than typical, with the intercept term coefficient being below the middle point of the Likert scale ($\beta = 4.555 \pm 0.470$, t = 9.688, $p < 0.001$).

We then compared this baseline model to models which contained fixed effect terms, in order to examine our second prediction that the type of construction a tool-use mention occurred in would impact how typical it was rated. First, the intercept-term only model was compared to a model with the number of times the tool appeared in a positive construction, total frequency of the tool, and their interaction as fixed effects. The fixed effects model provided a significantly better fit for the data ($\chi^2(3) = 36.299$, $p < 0.001$). Inspection of the model output suggested that, on average, an increase in positive count decreased the typicality rating given by participants ($\beta = -5.448\times10^{-01} \pm 2.191\times10^{-01}$, t = -2.487, p = 0.013). This supports our second prediction that tools mentioned in positive constructions are rated as less typical than those in negative constructions. The model suggests that an increase in total frequency increased the average typicality rating (β = 5.749e-01 $\pm$ 2.113e-01, t = 2.721, p = 0.007). The interaction between positive count and total frequency was not significant ($\beta = -1.360\times10^{-05} \pm 2.698\times10^{-04}$, t = -0.051, p = 0.960). As frequency increases, tools are rated as more typical, despite the general pattern that tools in these positive constructions were rated as more atypical overall.

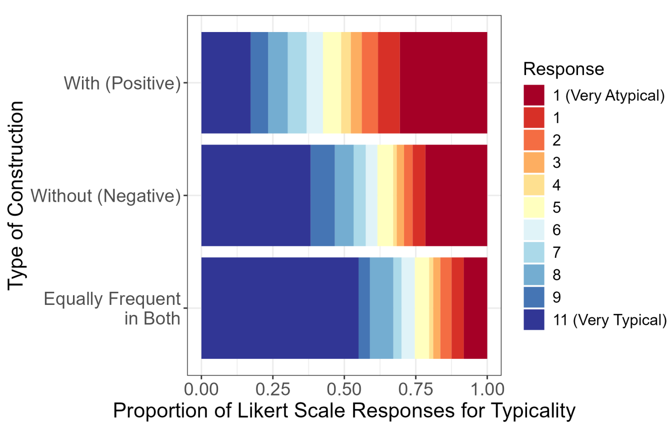

Figure 2: Stacked bar chart showing Likert responses by participants depending on what type of construction they were found in most frequently in the corpus data.

Second, an intercept-term only model was compared to a model with negative count and total frequency as fixed effects. Again, the fixed effects model provided a significantly better fit for the data ($\chi^2(3) = 38.299$, $p < 0.001$). Inspection of the model output suggested that an increase in negative count increased the typicality rating given by participants ($\beta = 7.923\times{}^{1}0{-}01 \pm 2.634\times{}^{1}0{-}01$, t = 3.008, p = 0.00269). This supports our second prediction that in general, items that were found on Reddit more often in a negative construction will be rated as more typical by participants. The model also suggests that total frequency increased average typicality rating ($\beta = 3.087\times{}^{1}0{-}2 \pm 6.137\times{}^{1}0{-}03$, t = 5.019, $p < 0.001$), as it did with the positive construction items. The interaction between negative count and total frequency was not significant ($\beta = -2.176\times{}^{1}0{-}02 \pm 1.487\times{}^{1}0{-}02$, t = -1.463, p = 0.144).

Finally, the interaction between positive and negative count was considered. A Likelihood Ratio Test showed that neither the negative nor positive fixed effects models provided a significant improvement in model fit over the other ($\chi^2(0) = 0$). A final model was fitted which directly tested the interaction between positive and negative count. Inspection of the model output suggests that the interaction between positive and negative count was significant and lead to a decrease in typicality rating ($\beta = -1.832 \pm 2.476\times{}^{1}0{-}01$ t = -7.398, p = 0). This interaction suggests that even for items that appeared frequently in negative constructions, if they also appeared frequently in positive constructions, they were more likely to be rated as atypical.

Discussion

Our study uses human ratings of VERB-DOBJ-TOOL triplets from Reddit data to investigate whether, in internet discourse, people tend to mention atypical tools more than typical tools in tool-use scenarios, in line with the `TELL IT LIKE IT ISN'T` theory that people are *more* inclined to mention newsworthy information.

Hypothesis 1: Ratings

Our results confirm our first prediction that tool-use scenarios from Reddit would be rated as more atypical by participants in our online task (e.g., `cut-food-ruler` or `clean-toilet-toothbrush`). This supports existing work in language production that shows speakers are more likely to mention properties of an event or object if the content being described is atypical (Mitchell et al., 2013). Reddit has hallmarks of being a community, such as diverse topics and self-reference (Singer et al., 2014). Thus, it is expected that parties within the community cooperate to facilitate the effective transfer of information, at an appropriate level of informativeness, during conversational exchanged (Grice, 1975). In this way, Reddit is functionally similar to a 'real-world' conversation, and so our headline result supports the conclusion that in ecologically natural situations people prefer to mention newsworthy or atypical tools. Although typicality was not directly manipulated and thus our results may not show the same reliability as an effect produced in a lab, it could be argued our naturally occurring data is ecologically valid by definition.

For many common actions (i.e., for VERB-DOBJ pairs which occurred more than once), the Reddit data contained a wide range of tools as the third item in the triplet, and these varied in the typicality ratings they received. A good example is shown by the total of 18 tools occurring for the `brush-teeth` verb and direct object pair. Of these 18, the generally accepted tool (toothbrush) is not explicitly mentioned in our data sample at all, and although several plausibly "enabling" tools were mentioned (toothpaste, bottled water, warm water, fingers), the rest seem very unlikely to be perceived as typical (brick, nailfile, razor, toilet brush, urine, wasabi).

Although we did not investigate effects of context in our study, in line with some (Glucksberg et al., 1986; Hess et al., 1995) albeit not all prior studies (Davies & Arnold, 2019; Nordmeyer & Frank, 2015; Troyer & McRae, 2021), the Reddit data preserves some surrounding words which gives some insight into their pragmatic purpose. What we find is that although the general trend is toward informative communication, some of the atypical tools are used in utterances that seem intended for hyperbolic or humorous effect. For example, "I bet when you wake up in the morning you brush your teeth with a razor", "Zarya is Russian and they brush their teeth with vodka", "I hope these players brush their teeth with the pavement". This faux-intimidating, bantering tone is found on the internet broadly (Steer et al., 2020; Whittle et al., 2019) but may be especially prevalent on Reddit. On one (cynical) hand, this bantering usage supports our typicality hypothesis. On the other, there seems a purer version of the hypothesis that only applies to literal usage: the concept of typicality seems more relevant to pure narratives of actions that actually happened than to jokes, insults, and idioms. These have their own conventions, e.g., it seems typical to exclaim, "Wash your mouth out with soap!" when someone swears, although the action itself may not be typical.

Hypothesis 2: With/without Constructions

The results show that, when blind to whether the items occurred originally in positive ("with") or negative ("without") constructions, people rate items that occurred in negative constructions as significantly more typical than items that occur in positive constructions. When people post on Reddit about performing actions *without* certain tools, the tools whose absences they have to work around would have been typical for the job. For example, the triplet `wash-clothing-soap` was rated highly typical and would presumably be accepted as normal by most people in the developed world, but occurred in text as, "I washed my clothing without soap in a stream, that was also my drinking water"; it is the fact the washing was performed *without* soap which makes the event worth telling.

One noticeable point is that the negative construction is often used in the data in a hypothetical sense, to express how undesirable it would be to perform an action without its typical tool. For example, the triplet `clean-dishes-hot_water` occurred as "How you clean dishes without hot water is another story...", and `heat-home-natural_gas` occurred as "I'm not sure how you could even heat your home without natural gas". These phrases express support for the normative behaviour expressed in the triplet, which were rated strongly typical by our participants. Like the hyperbolic 'banter' point above, this result supports our research question broadly construed, but on some level lacks the genuine evidence of atypicality that reports of behaviour would provide.

Frequency

For full ecological validity it is important to mention the effect of the frequency of each triplet on the total typicality rating, and why we decided not to include it. In the data for our experiment we necessarily ignored the naturally occurring frequency of each triplet (we selected triplets as long as they occurred at least twice and fulfilled the other criteria, but otherwise treated each the same and presented it to equal numbers of participants for rating). Under this design, hypotheses 1 and 2 are supported by our data and analyses.

However, were the frequency of the triplets to be accounted for, it may mitigate the atypicality effect isolated in our main experiment. Like many phenomena in linguistics, the Reddit data follows a broadly power-law distribution (cf. Zipf, 1935) (while there are very many atypical tools mentioned, each is only mentioned infrequently and even taken altogether they are outweighed by the more frequent and less diverse typical tools). A similar imbalance, echoing other corpora (Banjade & Rus, 2016; Konstantinova et al., 2012; Vincze et al., 2008) is that the Reddit data (and thus the experiment stimuli) contained far fewer instances of negative constructions than positive ones: of the c. 10,000 total participant ratings, c. 8,000 responses relate to positively-worded items, c. 1,300 responses to negative items, and a further c. 300 responses to items with equal positive and negative counts. The highest positive count for a single triplet within the stimuli data is 108, while the highest negative count is only 4. Thus, although high-frequency triplets are overwhelmingly positive constructions and positive construction generally increases people's atypicality rating, the frequency itself decreases it.

It can thus be seen that frequency *mediates* the effect of the positive/negative sentence construction. Sentence sign influences typicality through two causal paths: one direct, and one indirect through frequency. Thus although it is interesting to determine the effect of frequency, we therefore deem it unnecessary to include as a predictor when measuring the full effect of sentence sign on typicality rating, following Cinelli et al. (2024) who recommend to remove a mediator as a 'bad control' in this type of structure (see their Model 11).

NLP Implications

An important implication of our results is for application of natural language processing (NLP) systems. Great advances in recent years in these primarily neural networks (deep learning models) have consistently improved performance on language-use benchmarks. These systems are now often used in commercial settings like customer service or sales, or in traditionally-human arenas like translating or summarising text. However, as the systems learn about words using huge corpora of text taken from existing natural sources such as internet forums and social media, they learn the statistical relationships inherent in the sources. Concerns have already been raised about how NLP systems reproduce the biases and prejudices of normal human discourse and existing human society around gender (Zhao et al., 2019); hate speech (Luccioni & Viviano, 2021); and race (Blodgett & O'Connor, 2017). Our results raise the concern that NLP systems learn distorted ideas about the world and how it works, such as it being appropriate to `chop-carrots-axe` as well as `chop-carrots-knife`. Granted, we have not seen an outbreak of unusual language since the widespread introduction of large language models (although it could have been washed out by reinforcement learning by human feedback), but we still should be aware of the potential for inaccuracy if we assume that any snapshot of human language records the everyday physical habits of that society.

Limitations and Further Work

A number of limitations of the current work are related to the context and form of the data. Firstly, as we only looked at triplets in lemma form we could not test for any effect of tense or other morphology. However this was necessary to avoid any other confounding effects of negation (Just & Carpenter, 1971; Tian et al., 2010; Zwaan, 2012).

Secondly, our results rely on human ratings without accounting for previous findings that people comprehend typical constructions faster than atypical (Degen et al., 2015) or that comprehenders prefer atypicality (Rohde et al., 2021). However, we mitigated any difficulty that would be reflected in slower reaction times by letting participants take as long as they wanted to respond (within reason).

Finally, we drew our triplets from comments on posts rather than the original submitted posts (see Data). Despite no salient theoretical argument against our approach, it could be argued we did not study fully natural (i.e., spontaneous) discourse. However, as much natural language 'in the wild' is also produced in response to previous utterances, we consider Reddit comments functionally similar to natural discourse, although we broadly support further work to apply our method to original production of language.

Future work could test other language domains for the same effect, particularly numerical fields because quantitative metrics like average heights or temperatures have strong and available norms and may help avoid the effects of idiom and hyperbole.

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